

Reversal Variations Research I

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Abstract

We investigate five variations of a reversal strategy to address investors' tendency to overreact to short-term news. Using Barra asset data from 1996 to 2024, we compare standard return-based reversals with residualized approaches that isolate idiosyncratic movements. To mitigate the undesirable selection of stocks experiencing extreme negative news events, we implement winsorization and volume conditioning to limit exposure to tail outcomes. Our mean-variance optimal backtests with 5% active risk targeting reveal that while standard reversal achieves the highest information ratio over the full sample period (2000-2024), the enhanced signals outperform by approximately 2x in the out-of-sample period (2022-2024). These refinements successfully eliminate extreme allocations to distressed securities while maintaining strong risk-adjusted returns.

Sources

- *Evidence of Predictable Behavior of Security Returns*, 1990, Narasimhan and Jegadeesh
- *Persuasive Evidence of Market Inefficiency*, 1985, Rosenberg, Reid, and Lanstein

Economic Intuition

1. Standard Reversal

In the short term, investors tend to overreact to news, pushing the stock price either too high or too low relative to its fundamental value. This often reflects the cost of liquidity provision; traders who follow this strategy act as the buy or sell party for investors who are urgently trying to onload or offload their positions, thus earning a return for absorbing these temporary imbalances until the market corrects and reverts to the long-run mean.

2. Barra Reversal

The core intuition behind reversal is that investors overreact to news in the short term. The problem with using raw returns for a reversal signal is that they are affected by movements in systematic factors, such as the market, size, and value. By using the Barra specific return variable, we can calculate reversal using each firm's residualized returns.

3. Winsorized Barra Reversal

Because our optimizer gives high preference (weight) to high-reversal securities, the resulting portfolio allocates larger weights to stocks in the tails of the reversal signal distribution. This is undesirable, as stocks with extreme reversal scores (i.e., recent price plummets) often result from negative news events such as failing an FDA trial. We do not want to be tailing negative news, so by winsorizing the signal, we limit the impact of extreme reversal picks on our portfolio.

4. Volume-Adjusted Barra Reversal

A second approach to addressing the issues in our portfolio when picking extreme reversal stocks is to condition on trading volume. A stock's trading volume may signal news events when it exceeds typical levels. By reducing our alpha forecast when trading volume exceeds two standard deviations above its historical average, we eliminate the undesirable tendency of picking stocks with extreme negative news.

5. Winsorized Volume Adjusted Barra Reversal

While winsorizing and volume conditioning address the same issue with the reversal signals, their effects do not entirely overlap. We may therefore construct a more robust reversal signal by combining both methodologies.

Data

For our research, we use the Barra assets data available through sf-quant. Note that this data is point-in-time compliant and completely tradable.

Signal Construction

1. Standard Reversal

The standard signal is constructed by taking the daily log returns of each stock and performing a rolling sum over a 21-day window. This value is then multiplied by -1.

2. Barra Reversal

The residualized reversal signal is constructed by taking the daily specific returns of each stock and performing an exponential weighted moving average with a span of 5. This value is then multiplied by -1 and lagged by 1 trading day.

3. Winsorized Barra Reversal

After computing the cross-sectional Barra reversal score, we clip it to the range -2 to +2. The alpha calculation then uses these adjusted reversal z-scores.

4. Volume Adjusted Barra Reversal

For each day, we first calculate the time series z-score of each stock's dollar trading volume over the past 252 trading days. We next set the stock's alpha value to 0 if its cross-sectional reversal z-score is extreme (< -2 or > 2) and the time series volume z-score is above 2.

5. Winsorized Volume Adjusted Barra Reversal

The previous two methods are combined by first winsorizing the cross-sectional reversal z-score. The subsequent alpha adjustment is the same, except that extreme reversal is capped at the -2 and +2 levels.

Portfolio Results

For each signal, we find the optimal portfolio for the last trading day (2025-12-31). We use the full constraint set and target an active risk of 5%. A \$5 price filter is applied prior to signal scoring. Since all signals are derived from the same core signal, the backtest plots are mostly

identical and have been omitted accordingly. All plots are available in the GitHub repo, and an interactive Marimo notebook is provided for comparing each strategy across different sample periods. The following is commentary on the top 10 positions in each portfolio.

1. **Standard Reversal**

A sample portfolio for this signal is shown in Figure 2. The systematic bets against the benchmark through active weighting are shown through plays such as taking a significant active position in NVDA (-17.29% benchmark change). However, this signal is subject to extreme outliers and negative news. This can be seen in the signal picking up a “high conviction” play in JANX, a therapeutics company whose stock dropped sharply following drug test results.

2. **Barra Reversal**

As shown in Figure 5, the Barra reversal portfolio actually exhibits stronger reversal allocation than the standard reversal portfolio. This is due to the signal's residualized nature, which allows it to pick up solely on idiosyncratic movements in the underlying. The most interesting position is RARE with a 10% total allocation. RARE is a biotech company whose stock recently declined sharply.

3. **Winsorized Barra Reversal:** As shown in Figure 8, winsorizing effectively eliminates the most extreme weight shifts in the portfolio. We no longer pick up adverse biotech stocks. However, we still have large weight shifts for other companies, such as Coca-Cola.

4. **Volume Adjusted Barra Reversal:** Figure 10 shows the resulting portfolio for this method. We find that the portfolio is close to that of the winsorizing method, validating the intuition behind both methodologies.

5. **Winsorized Volume-Adjusted Barra Reversal:** After combining both reversal enhancements, Figure 12 shows that the portfolio no longer exhibits extreme weight changes.

Backtest Results

We perform Mean-Variance Optimal backtests for each signal. We constrain the portfolios to have a beta of zero relative to the market and to have the portfolio weights sum to zero. A \$5 price filter is applied prior to the portfolio construction step. For each signal, we pick a gamma that achieves approximately 5% active risk over the period from 1996-07-31 to 2024-12-31. After computing the weights for every trading day in the full sample, we use the weights and forward returns for the 2000-2024 (full sample) and 2022-2024 (out-of-sample) periods to highlight the performance of each signal.

Overall, we find that the standard reversal signal has the highest Sharpe ratio (i.e., information ratio, since we're in an active space) for the full sample period. When we limit ourselves to just the out-of-sample period, we actually find that the derivative reversal signals are all about two times better than the standard reversal signal. In general, the performance of each strategy is

very strong, and no signal is especially distinct from the others. While standard reversal is the best strategy in the full sample, we believe it would actually incur the most transaction costs, since it captures large allocations in extreme reversals and selects securities known to have lower liquidity.

Appendix

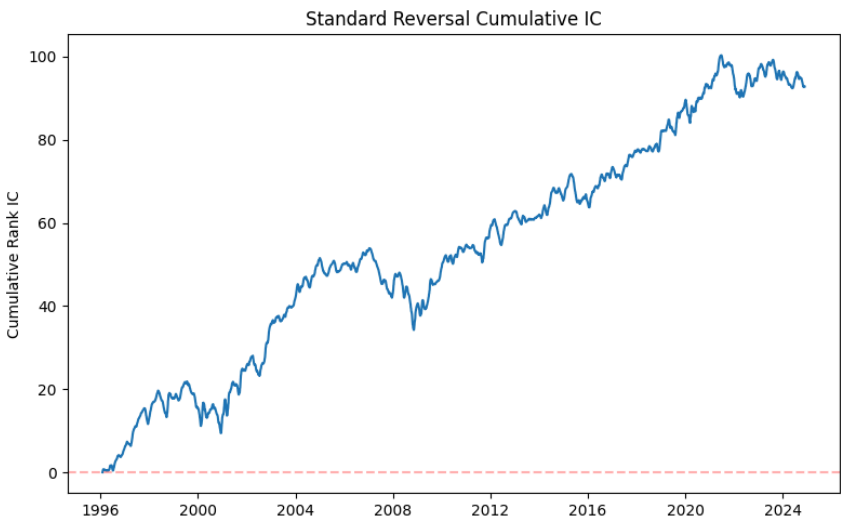
1. Reversal Signal Construction

```
def reversal() → pl.Expr:
    return (
        pl.col('return')
        .log1p()
        .rolling_sum(21)
        .mul(-1)
        .over('barrid')
        .alias('reversal')
    )
```

2. Reversal Sample Portfolio

Standard Reversal Portfolio						
Date	Ticker	barrid	Total Weight	Benchmark Weight	Active Weight	Benchmark Percent Change
2025-12-30	AAPL	USAB1X1	6.04%	5.78%	0.27%	4.64%
2025-12-30	MSFT	USAJ471	5.63%	5.19%	0.44%	8.55%
2025-12-30	NVDA	USA2HB1	5.40%	6.52%	-1.13%	-17.29%
2025-12-30	GOOG	USA1LI3	5.25%	2.43%	2.81%	115.60%
2025-12-30	GOOGL	USA1LI1	5.17%	2.61%	2.55%	97.64%
2025-12-30	AMZN	USAXO31	3.42%	3.56%	-0.14%	-3.93%
2025-12-30	MA	USA39K1	3.14%	0.74%	2.40%	325.99%
2025-12-30	AVGO	USAAUE1	2.88%	2.37%	0.51%	21.47%
2025-12-30	JANX	USBFQX1	2.71%	0.00%	2.71%	227,472.95%
2025-12-30	BRK.B	USABPA2	2.16%	0.99%	1.17%	118.52%

3. Reversal Cumulative Information Coefficient



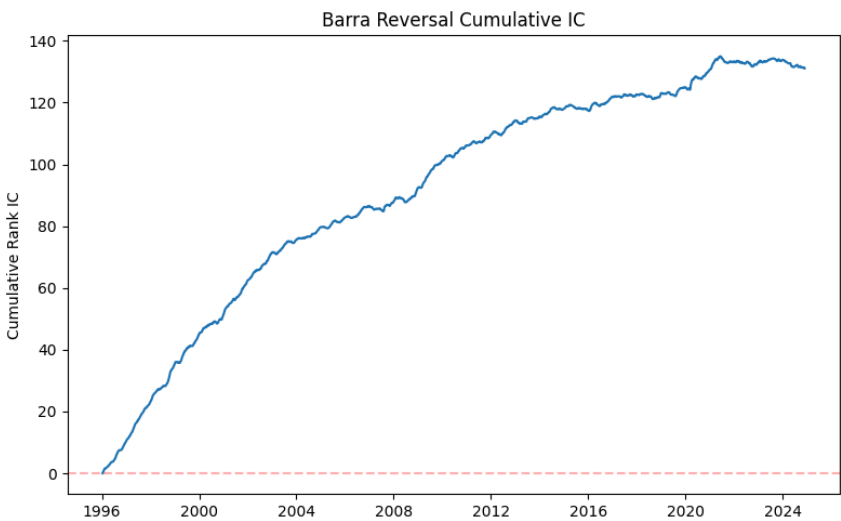
4. Barra Residual Reversal Signal Construction

```
def barra_reversal() → pl.Expr:
    return (
        pl.col("specific_return")
        .ewm_mean(span=5, min_samples=5)
        .mul(-1)
        .shift(1)
        .over("barrid")
        .alias("barra_reversal")
    )
```

5. Barra Residual Reversal Sample Portfolio

Barra Reversal Portfolio						
Date	Ticker	barrid	Total Weight	Benchmark Weight	Active Weight	Benchmark Percent Change
2025-12-30	RARE	USBBOD1	10.63%	0.00%	10.63%	337,842.13%
2025-12-30	MSFT	USAJ471	5.87%	5.19%	0.68%	13.09%
2025-12-30	NVDA	USA2HB1	5.47%	6.52%	-1.06%	-16.22%
2025-12-30	AAPL	USAB1X1	5.46%	5.78%	-0.32%	-5.55%
2025-12-30	GOOG	USA1LI3	5.23%	2.43%	2.80%	114.96%
2025-12-30	GOOGL	USA1LI1	4.98%	2.61%	2.37%	90.54%
2025-12-30	AMZN	USAXO31	2.93%	3.56%	-0.62%	-17.54%
2025-12-30	MA	USA39K1	2.92%	0.74%	2.18%	296.42%
2025-12-30	L	USAIBJ1	2.64%	0.03%	2.61%	8,299.26%
2025-12-30	COKE	USAD221	2.13%	0.01%	2.12%	16,906.05%

6. Barra Residual Reversal Cumulative Information Coefficient



7. Winsorized Barra Residual Reversal Signal Construction

```
def barra_reversal_score() → pl.Expr:
    return (
        pl.col("barra_reversal")
        .sub(pl.col("barra_reversal").mean())
        .truediv(pl.col("barra_reversal").std())
        .over("date")
        .alias("barra_reversal_score")
    )

def winsorized_barra_reversal() → pl.Expr:
    return (
        pl.col("barra_reversal_score")
        .clip(
            lower_bound=-2,
            upper_bound=2,
        )
        .over("date")
        .alias("winsorized_barra_reversal_score")
    )
```

8. Winsorized Barra Residual Reversal Sample Portfolio

Winsorized Barra Reversal Portfolio						
Date	Ticker	barrid	Total Weight	Benchmark Weight	Active Weight	Benchmark Percent Change
2025-12-30	MSFT	USAJ471	5.54%	5.19%	0.36%	6.89%
2025-12-30	AAPL	USAB1X1	5.14%	5.78%	-0.64%	-11.07%
2025-12-30	NVDA	USA2HB1	4.71%	6.52%	-1.82%	-27.86%
2025-12-30	GOOG	USA1LI3	3.92%	2.43%	1.49%	61.19%
2025-12-30	MA	USA39K1	3.49%	0.74%	2.75%	373.61%
2025-12-30	GOOGL	USA1LI1	3.45%	2.61%	0.84%	32.06%
2025-12-30	COKE	USAD221	3.21%	0.01%	3.20%	25,558.38%
2025-12-30	JNJ	USAHML1	3.19%	0.71%	2.48%	347.13%
2025-12-30	BRK.B	USABPA2	2.71%	0.99%	1.72%	173.73%
2025-12-30	L	USAIBJ1	2.37%	0.03%	2.34%	7,448.00%

9. Volume Adjusted Barra Residual Reversal Signal Construction

```
def dollar_volume_score() → pl.Expr:
    return (
        pl.col("dollar_volume")
        .sub(pl.col("dollar_volume").rolling_mean(window_size=252, min_samples=1))
        .truediv(pl.col("dollar_volume").rolling_std(window_size=252, min_samples=2))
        .over("barrid")
        .alias("dollar_volume_score")
    )

def volume_adjusted_barra_reversal() → pl.Expr:
    return (
        pl.when(
            pl.col("barra_reversal_score").gt(2) & pl.col("dollar_volume_score").gt(2)
        )
        .then(pl.lit(0))
        .otherwise(pl.col("barra_reversal_score"))
        .alias("volume_adjusted_barra_reversal_score")
    )
```

10. Volume Adjusted Barra Residual Reversal Sample Portfolio

Volume Conditioned Barra Reversal Portfolio						
Date	Ticker	barrid	Total Weight	Benchmark Weight	Active Weight	Benchmark Percent Change
2025-12-30	MSFT	USAJ471	5.73%	5.19%	0.55%	10.56%
2025-12-30	AAPL	USAB1X1	5.33%	5.78%	-0.45%	-7.79%
2025-12-30	NVDA	USA2HB1	5.12%	6.52%	-1.41%	-21.58%
2025-12-30	GOOG	USA1LI3	4.59%	2.43%	2.16%	88.67%
2025-12-30	GOOGL	USA1LI1	4.26%	2.61%	1.64%	62.88%
2025-12-30	MA	USA39K1	3.49%	0.74%	2.76%	374.01%
2025-12-30	JNJ	USAHML1	2.96%	0.71%	2.25%	314.86%
2025-12-30	COKE	USAD221	2.73%	0.01%	2.72%	21,703.63%
2025-12-30	L	USAIBJ1	2.47%	0.03%	2.44%	7,768.28%
2025-12-30	BRK.B	USABPA2	2.41%	0.99%	1.42%	143.20%

11. Winsorized Volume Adjusted Barra Residual Reversal Signal Construction

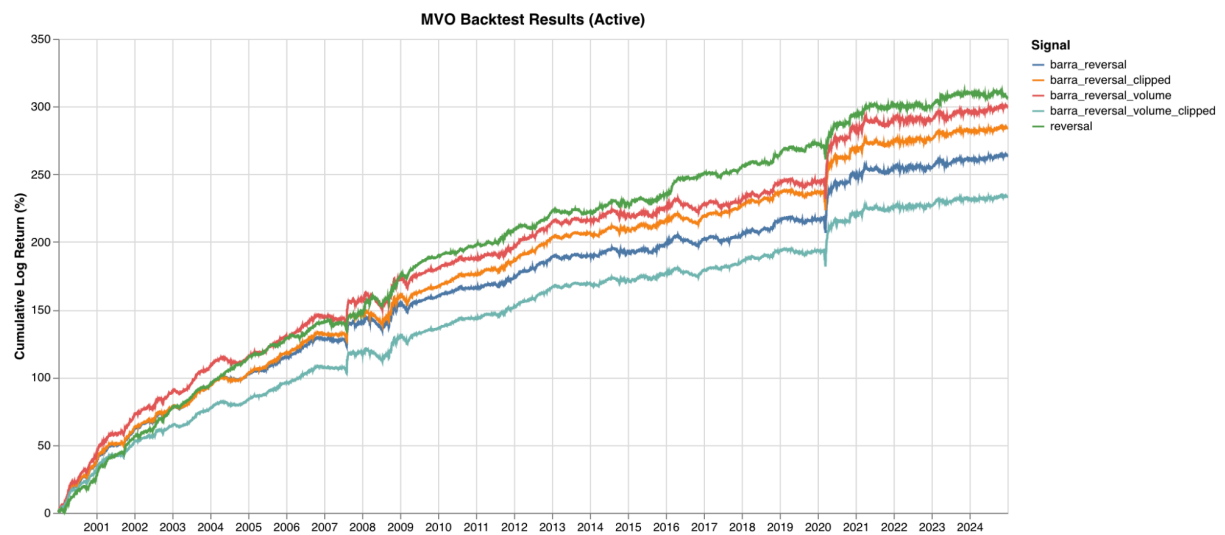
```
def winsorized_barra_reversal() → pl.Expr:
    return (
        pl.col("barra_reversal_score")
        .clip(
            lower_bound=-2,
            upper_bound=2,
        )
        .over("date")
        .alias("winsorized_barra_reversal_score")
    )

def winsorized_volume_adjusted_barra_reversal() → pl.Expr:
    return (
        pl.when(
            pl.col("winsorized_barra_reversal_score").gt(2)
            & pl.col("dollar_volume_score").gt(2)
        )
        .then(pl.lit(0))
        .otherwise(pl.col("barra_reversal_score"))
        .alias("winsorized_volume_adjusted_barra_reversal_score")
    )
```

12. Winsorized Volume Adjusted Barra Residual Reversal Sample Portfolio

Volume Conditioned Barra Reversal Portfolio						
Date	Ticker	barrid	Total Weight	Benchmark Weight	Active Weight	Benchmark Percent Change
2025-12-30	MSFT	USAJ471	5.48%	5.19%	0.30%	5.71%
2025-12-30	AAPL	USAB1X1	5.36%	5.78%	-0.42%	-7.27%
2025-12-30	NVDA	USA2HB1	5.25%	6.52%	-1.27%	-19.45%
2025-12-30	GOOG	USA1LI3	4.38%	2.43%	1.94%	79.82%
2025-12-30	GOOGL	USA1LI1	4.06%	2.61%	1.45%	55.31%
2025-12-30	MA	USA39K1	3.11%	0.74%	2.37%	322.01%
2025-12-30	BRK.B	USABPA2	2.83%	0.99%	1.84%	186.12%
2025-12-30	AMZN	USAXO31	2.56%	3.56%	-1.00%	-28.03%
2025-12-30	JNJ	USAHML1	2.39%	0.71%	1.67%	234.22%
2025-12-30	META	USAMZ01	2.29%	2.08%	0.22%	10.52%

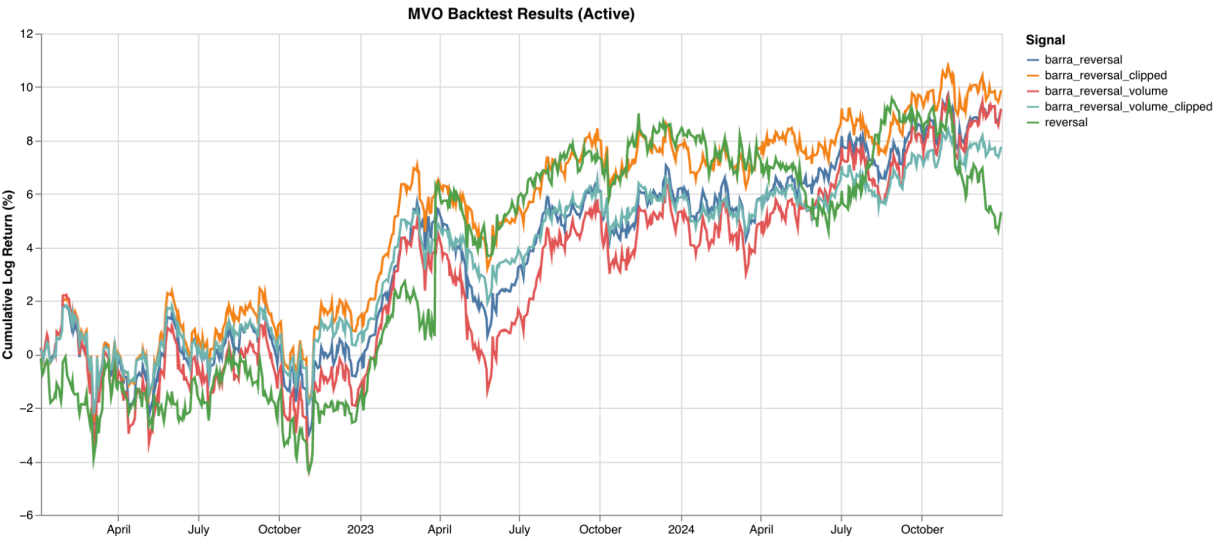
13. Full Sample (2000-2024) Backtest Chart



14. Full Sample (2000-2024) Backtest Summary Table

MVO Backtest Results (Active)			
Signal	Mean Return	Volatility	Sharpe
barra_reversal	10.72%	4.99%	2.15
barra_reversal_clipped	11.53%	5.20%	2.22
barra_reversal_volume	12.22%	5.99%	2.04
barra_reversal_volume_clipped	9.47%	4.43%	2.14
reversal	12.43%	4.88%	2.55

15. Out of Sample (2022-2024) Backtest Chart



16. Out of Sample Backtest Summary Table

MVO Backtest Results (Active)			
Signal	Mean Return	Volatility	Sharpe
barra_reversal	3.18%	4.90%	0.65
barra_reversal_clipped	3.41%	4.85%	0.70
barra_reversal_volume	3.23%	5.89%	0.55
barra_reversal_volume_clipped	2.67%	4.16%	0.64
reversal	1.95%	6.05%	0.32